

Introduction:

Good Morning to all of you,

Myself ALLAM SRIDHAR, am here with my team members Venkatesh Paturu and Haseeb are here to present our Dashboard/Analysis results.

This dashboard was created to provide Excelerate with a comprehensive view of their user engagement and opportunity completion across various demographics, locations, and skills. The data reflects insights into the platform's user base and program effectiveness, helping Excelerate understand both who is using the platform and how they are engaging with it.

The dashboard consolidates data from two primary sources:

- **User Data** - contains details of all registered users, gives us broad view of user base demographics
- **Opportunity Sign-Up and Completion Data** - focused on users engaged in specific opportunities, detailing their sign-ups, completions, and achievements.

Together, these datasets offer insights into user demographics, regional engagement, popular opportunities, and outcomes like completed opportunities and scholarships awarded.

Some decisions which we considered in preprocessing were like :

- In userdataset there were around 9535 null values in Gender column, 62 null values in Country column, 10812 null values in Degree Column, 9534 null values in city column, 9544 null values in zip column, 9 null values in isFromSocialMedia column.
- So instead of dropping these columns, since we will be losing lots of information if we do so, we opted different strategies like for Degree column we replaced null values with Not mentioned, In gender column we replaced null values with Not Mentioned (here we didn't replace with most frequent value is because in this data set there is not much useful analysis can be done except for demographic distribution which includes gender distribution, so if we imputed with most frequent value then the ratio might have been biased), Now for IsFromSocialMedia column since only 9 values are missing we are replacing with most frequent value. But for city and zip column we are grouping countries together and finding out most frequent value and replacing these with null values.
- In Opportunities dataset there were 1 missing values in state, zipcode, current/Intensed major, city and state column so we dropped them, then for columns like Skill points Earned, Reward Amount, Badge Id, Badge Name, Skills Earned columns we assumed that these columns contained values only if the candidate has successfully completed the opportunity, else null values were present, so instead of going for imputing most frequent value, we replaced null values with 0 in numerical columns like Reward Amount and Skill Points Earned. Then for missing values in Badge Id, Badge Name and skills Earned we replaced them with NaN itself, Indicating opportunity not completed yet.

Now let's move to dashboard and understand the analysis:

1st slide:

Q1. How many people are signed up on the platform, and how many of those have signed up for opportunities?

- We can see that we do have 27500 total signups, only 11,435 users signing up for opportunities, Opportunity signup percentage is 41.2%, there's an engagement gap of 58.42%. This reveals a substantial portion of the user base who may need further motivation or clarity about the platform's offerings to participate in opportunities.
- Excelerate can leverage this data to design targeted engagement campaigns. For instance, email outreach, onboarding tutorials, or notifications that showcase the benefits of signing up for opportunities could effectively re-engage inactive users.

Q2. Top 10 Countries Learners Have Signed Up From?

- The substantial user base in India (11,893), Nigeria (4,357), and the United States (3,691) suggests an opportunity to increase marketing efforts in these regions.
- Excelerate could create targeted campaigns that speak to each country's specific professional trends, cultural elements, or industry needs, helping increase relevance and engagement.
- Excelerate could develop partnerships with organizations or educational institutions within these countries to offer internships, projects, or mentorships that align with local industry demands.
- **Targeted Development Initiatives:**
With Egypt, Ghana, and Bangladesh contributing significantly, Excelerate might consider customized development initiatives that align with these countries' prevalent industries or educational gaps. Offering specialized courses or skill development programs in areas relevant to the job markets in these countries could make the platform a valuable career asset for users there.

Slide 2:

Q3. Cities in the US Learners Have Signed Up From

- With the majority of sign-ups from St. Louis (2,114), followed by Chicago, Excelerate can consider creating city-specific events or partnerships. For example, partnerships with local organizations, educational institutions, or career fairs in St. Louis and Chicago could increase visibility and encourage further sign-ups or completions.
- Understanding that many users are graduates (1,300+), high school students (692), and undergraduates (434), Excelerate can offer programs specifically designed for different educational levels. For graduates, this could mean advanced certifications or job placement assistance, while for high school students, it could involve introductory career programs or internships that align with their future study or career interests.
- With peak sign-ups occurring from June to August, Excelerate could maximize its reach by launching targeted summer campaigns or seasonal workshops. By aligning promotions or new offerings with this period, Excelerate can capitalize on the higher interest levels during these months.

Slide 3:

Q4. Most Popular Opportunity Learners Have Signed Up For

- Data Visualization (28%), Project Management (19.2%), and Digital Marketing (12.2%) are the most popular opportunities, making up nearly 60% of all sign-ups. Excelerate can leverage this trend by expanding content in these areas, such as offering advanced courses, certification tracks, or hands-on projects that build on foundational skills.
- With a completion ratio of 12.41% (2,514 completions out of 20,257 sign-ups), there's an opportunity to investigate barriers learners face in completing opportunities. Excelerate could consider:
 - Reminders and check-ins for learners enrolled in high-demand courses, such as Data Visualization and Project Management.
 - Modular course options to allow users to progress in smaller, more manageable segments.
 - Completion incentives, such as badges, certificates, or professional credits, to motivate learners.
- With a gender distribution of 60% male and 39% female, Excelerate might explore initiatives to encourage female engagement in high-interest fields like data visualization and project management, which can sometimes be perceived as male-dominated. This might include women-focused mentorship programs or scholarships specifically for women.

Q5. Most Popular Opportunity Learners Have Completed

- Data Visualization leads with 33% of completed opportunities, followed by Project Management (17.8%) and Digital Marketing (10.3%). This shows that learners are most inclined to complete courses that offer clear, technical skill-building. Expanding advanced modules in these areas could cater to this demand, enabling learners to progress within a defined skill track.
- The completion gender distribution shows 57% male and 42% female, reflecting an improved completion rate among females compared to their initial sign-up share. Excelerate can build on this trend by promoting content or mentorship specifically aimed at supporting female learners, especially in popular fields like project management and healthcare management, to sustain and grow female engagement
- With a completion ratio of 12.41% across the board, Excelerate could implement interactive reminders or progress milestones to encourage users to reach completion. This could particularly target programs with lower completion rates, reinforcing the value of finishing courses.

Slide 4:

Q6. Demographics of Those Who Have Signed Up and Completed (Gender, Student Status, etc.)

Although we did cover this in earlier slides, but let us see them again,

- Total opportunity signups 20,257, completed opportunities 2,514, Total male signups are: 60.3% (i.e. 12,207) out of which completion rate is 11.76%, total female are 39.4%(i.e. 7973) out of which completion rate is 13.32%.

- Female learners have a higher completion rate (13.4%) compared to male learners (11.7%). This indicates a strong engagement and commitment among female participants, suggesting that Excelerate could expand offerings or provide additional support tailored for female learners, particularly in fields where they show higher engagement.
- These similar completion rates across educational backgrounds reflect Excelerate's broad appeal. However, high school students and individuals not in formal education have slightly higher completion rates, indicating an interest among early learners and non-traditional students. Excelerate could capitalize on this by designing introductory or foundational courses that align with high school and non-formal education needs, potentially fostering long-term platform loyalty.
- Given that high school students and those not currently in education have shown a strong completion rate, these segments represent a valuable audience for Excelerate. Developing pathways or early-career tracks could better serve these groups, providing them with opportunities for skill-building and career preparation.

Slide 5:

Q7. Most Gained Skills on Excelerate

- This data emphasizes learners' desire to build soft skills alongside digital literacy, which is crucial in today's job market. Excelerate could consider creating more targeted programs or specialized tracks that cater to these top skills.
- Total 14 unique skills and total skills count is 14,236
- The emphasis on critical thinking, communication, and collaboration aligns well with Excelerate's goal of fostering employability through foundational skill development.
- Addressing diverse skill needs across gender categories and potentially creating gender-sensitive programs may further increase engagement and completion rates.

Note: In **preferred Sponsors** , we can see that the distribution is almost same just a little difference.

- This data highlights sponsor preferences that Excelerate could leverage to **forge deeper partnerships** or co-develop content with these popular sponsors.
- Expanding collaborations with high-preference sponsors or involving new, similar institutions in key growth regions could diversify offerings and expand Excelerate's appeal across global learner segments.

Slide 6:

Q8. Total Scholarship Awarded and Through Which Opportunities

- A substantial total scholarship amount of **\$2,716,860** has been awarded across various opportunities, reflecting Excelerate's commitment to enhancing access to education and skill development for learners.
- A total of **2,514 scholarships** have been distributed, showcasing the platform's role in providing financial assistance to learners seeking opportunities for growth.
- **Male Learners:** Awarded **\$1.6 million**, **Female Learners:** Received **\$1.2 million**, **Don't Want to Specify:** **\$5,100** and **Others:** **\$3,000**

- The substantial scholarship amounts in Data Visualization, Project Management, and Digital Marketing, Health care Management indicate strong interest areas. Excelerate should consider expanding offerings in these domains to accommodate growing demand.
- Opportunities like the Epidemiology Training Internship and Slide Geeks competition, though receiving lower total awards, represent unique fields that could benefit from increased visibility. Enhanced marketing efforts could drive more interest and participation.
- The average scholarship amount can serve as a benchmark for future financial planning and allocation. Understanding which programs offer higher average scholarships can help Excelerate prioritize resources effectively.

Slide 7:

- Current/Intended major indicates the field which the user belongs to or instrested in. Here we can see that most of the Top 10 Majors are related to Computers background, out of which most of them have opted for Data visualization and Project Management Opportunity.
- Here the Top 6 earned badges belongs to these Data visualizations and Project Management Opportunities only.

Slide 8:

Trend Analysis

Apply Date

- Actually it is like the applying for an opportunity on so and so dates
- If we see yearly distribution, clearly we can see that there is tremendous growth in number of applications like from 701 to 19,555
- If we see monthly distribution June to August shows highest amount of applications
- This trend is similar across all the genders, so there is no gender based trend analysis here

Opportunity Start Date

- Actually it refers to when the user has started the opportunity i.e. the start date of the Opportunity.
- If we see yearly Trend/ Distribution, we can clearly see that From 2022 to 2023, there is a tremendous growth like from 1,666 to 16,336. In 2024 it is 2,255 (please note that it has not dropped, but the point is that we only have 1 month of data of this year)
- If we see monthly trend we can see that there are spikes in the months of November, March and from June to August. (Again in Jan 2024)

Opportunity End Date

- Actually it refers to when the opportunity will be ending, i.e. the Deadline of the Opportunity
- If we see yearly Trend/ Distribution, we can clearly see that From 2022 to 2024 the Opportunity End Date count is being increasing, i.e more number of opportunities are being every year compared to previous year.
- If we see monthly trend, some opportunities are ending in Aug 2023, Most of the opportunities are ending in Jan 2024.